

Q1 (2)

$$x \in \left(0, \frac{\pi}{2}\right)$$

$$\log_{10} \sin x + \log_{10} \cos x = -1$$

$$\Rightarrow \log_{10} \sin x \cdot \cos x = -1$$

$$\Rightarrow \sin x \cdot \cos x = \frac{1}{10}$$

$$\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$$

$$\Rightarrow \sin x + \cos x = 10^{\left(\log_{10} \sqrt{n} - \frac{1}{2}\right)} = \sqrt{\frac{n}{10}}$$

by squaring

$$1 + 2 \sin x \cdot \cos x = \frac{n}{10}$$

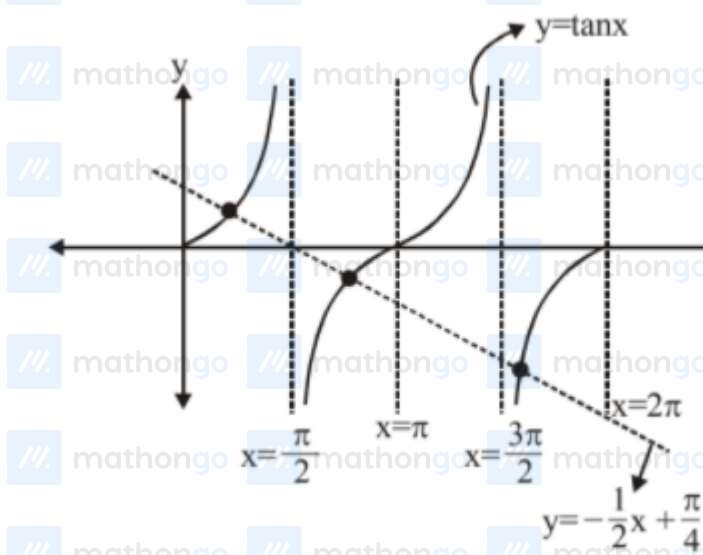
$$\Rightarrow 1 + \frac{1}{5} = \frac{n}{10} \Rightarrow n = 12$$

Q2 (1)

$$x + 2 \tan x = \frac{\pi}{2}$$

$$\Rightarrow 2 \tan x = \frac{\pi}{2} - x$$

$$\Rightarrow \tan x = -\frac{1}{2}x + \frac{\pi}{4}$$



Number of solutions of the given equation is '3'.

Q3 (1)

If $\cot x > 0 \Rightarrow \frac{1}{\sin x} = 0$ (Not possible)

If $\cot x < 0 \Rightarrow 2 \cot x + \frac{1}{\sin x} = 0$

$\Rightarrow 2 \cos x = -1$

$\Rightarrow x = \frac{2\pi}{3}$ or $\frac{4\pi}{3}$ (reject)

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

If for $x \in \left(0, \frac{\pi}{2}\right)$, $\log_{10} \sin x + \log_{10} \cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, $n > 0$ then the value of n is equal to :

(1) 20

(2) 12

(3) 9

(4) 16

Q2: 17 March (Shift 2) - Single Correct

The number of solutions of the equation

$x + 2 \tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is :

(1) 3

(2) 4

(3) 2

(4) 5

Q3: 18 March (Shift 1) - Numerical

The number of solutions of the equation $|\cot x| = \cot x + \frac{1}{\sin x}$ in the interval $[0, 2\pi]$ is

Answer Key

Q1 (2)

Q2 (1)

Q3 (1)